MSG No. 200.16 $Pm'\bar{3}'$ [Type III, cubic]

Table 1: Wyckoff site: `1a`, site symmetry: `m'-3'`.

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `1b`, site symmetry: `m'-3'`.

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `3c`, site symmetry: `m'm'm'..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,8,10,12,17,20,22,24]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,7,9,11,18,19,21,23]`

Table 4: Wyckoff site: `3d`, site symmetry: `m'm'm'..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[5,8,10,12,17,20,22,24]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[6,7,9,11,18,19,21,23]`

Table 5: Wyckoff site: `6e`, site symmetry: `2m'm'..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2,15,16]`
2	$[-x, 0, 0]$	`[3,4,13,14]`
3	$[0, x, 0]$	`[5,12,20,22]`
4	$[0, 0, x]$	`[6,9,19,23]`
5	$[0, 0, -x]$	`[7,11,18,21]`
6	$[0, -x, 0]$	`[8,10,17,24]`

Table 6: Wyckoff site: `6f`, site symmetry: `2m'm'..`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,2,15,16]`
2	$\left[-x, 0, \frac{1}{2}\right]$	`[3,4,13,14]`
3	$\left[\frac{1}{2}, x, 0\right]$	`[5,12,20,22]`
4	$\left[0, \frac{1}{2}, x\right]$	`[6,9,19,23]`
5	$\left[0, \frac{1}{2}, -x\right]$	`[7,11,18,21]`

continued ...

Table 6

No.	position	mapping
6	$\left[\frac{1}{2}, -x, 0\right]$	[8,10,17,24]

Table 7: Wyckoff site: 6g, site symmetry: 2m'm'..

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,2,15,16]
2	$\left[-x, \frac{1}{2}, 0\right]$	[3,4,13,14]
3	$\left[0, x, \frac{1}{2}\right]$	[5,12,20,22]
4	$\left[\frac{1}{2}, 0, x\right]$	[6,9,19,23]
5	$\left[\frac{1}{2}, 0, -x\right]$	[7,11,18,21]
6	$\left[0, -x, \frac{1}{2}\right]$	[8,10,17,24]

Table 8: Wyckoff site: 6h, site symmetry: 2m'm'..

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,15,16]
2	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[3,4,13,14]
3	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[5,12,20,22]
4	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[6,9,19,23]
5	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[7,11,18,21]
6	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[8,10,17,24]

Table 9: Wyckoff site: 8i, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,5,6]
2	$[x, -x, -x]$	[2,10,11]
3	$[-x, x, -x]$	[3,7,12]
4	$[-x, -x, x]$	[4,8,9]
5	$[-x, -x, -x]$	[13,17,18]
6	$[-x, x, x]$	[14,22,23]
7	$[x, -x, x]$	[15,19,24]
8	$[x, x, -x]$	[16,20,21]

Table 10: Wyckoff site: 12j, site symmetry: m'..

No.	position	mapping
1	$[0, y, z]$	[1,14]
2	$[0, -y, -z]$	[2,13]
3	$[0, y, -z]$	[3,16]
4	$[0, -y, z]$	[4,15]
5	$[z, 0, y]$	[5,24]

continued ...

Table 10

No.	position	mapping
6	$[y,\ z,\ 0]$	`[6,21]`
7	$[-y,\ z,\ 0]$	`[7,23]`
8	$[-z,\ 0,\ y]$	`[8,22]`
9	$[-y,\ -z,\ 0]$	`[9,18]`
10	$[z,\ 0,\ -y]$	`[10,20]`
11	$[y,\ -z,\ 0]$	`[11,19]`
12	$[-z,\ 0,\ -y]$	`[12,17]`

Table 11: Wyckoff site: `12k`, site symmetry: `m'..`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ z\right]$	`[1,14]`
2	$\left[\frac{1}{2},\ -y,\ -z\right]$	`[2,13]`
3	$\left[\frac{1}{2},\ y,\ -z\right]$	`[3,16]`
4	$\left[\frac{1}{2},\ -y,\ z\right]$	`[4,15]`
5	$\left[z,\ \frac{1}{2},\ y\right]$	`[5,24]`
6	$\left[y,\ z,\ \frac{1}{2}\right]$	`[6,21]`
7	$\left[-y,\ z,\ \frac{1}{2}\right]$	`[7,23]`
8	$\left[-z,\ \frac{1}{2},\ y\right]$	`[8,22]`
9	$\left[-y,\ -z,\ \frac{1}{2}\right]$	`[9,18]`
10	$\left[z,\ \frac{1}{2},\ -y\right]$	`[10,20]`
11	$\left[y,\ -z,\ \frac{1}{2}\right]$	`[11,19]`
12	$\left[-z,\ \frac{1}{2},\ -y\right]$	`[12,17]`

Table 12: Wyckoff site: `24l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x,\ -y,\ -z]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$[y,\ z,\ x]$	`[6]`
7	$[-y,\ z,\ -x]$	`[7]`
8	$[-z,\ -x,\ y]$	`[8]`
9	$[-y,\ -z,\ x]$	`[9]`
10	$[z,\ -x,\ -y]$	`[10]`
11	$[y,\ -z,\ -x]$	`[11]`
12	$[-z,\ x,\ -y]$	`[12]`
13	$[-x,\ -y,\ -z]$	`[13]`
14	$[-x,\ y,\ z]$	`[14]`
15	$[x,\ -y,\ z]$	`[15]`
16	$[x,\ y,\ -z]$	`[16]`
17	$[-z,\ -x,\ -y]$	`[17]`

continued ...

Table 12

No.	position	mapping
18	$[-y, -z, -x]$	[18]
19	$[y, -z, x]$	[19]
20	$[z, x, -y]$	[20]
21	$[y, z, -x]$	[21]
22	$[-z, x, y]$	[22]
23	$[-y, z, x]$	[23]
24	$[z, -x, y]$	[24]